30 A detector of ionising radiation gives a background count rate of 24 per minute.

A radioactive source is placed close to the detector and the reading is 532 counts per minute. What will be the reading after two half-lives of the source?

A 127

B 133

C 151

D 157

Ans: C

Count rate due to source now = 532 – 24 = 508.

Count rate due to source 2 half-lives later = 508 /4 = 127.

Count rate due to source and background = 127 + 24 = 151.